

BCA-205 OPERATING SYSTEM & UNIX

Introduction: Introduction to Operating Systems, Operating system services, multiprogramming, time-sharing system, storage structures, system calls, multiprocessor system. Basic concepts of CPU scheduling, Scheduling criteria, Scheduling algorithms, algorithm evaluation, multiple processor scheduling, real time scheduling I/O devices organization, I/O devices organization, I/O devices organization, I/O buffering.

Process Management: Process concept, process scheduling, operations on processes, threads, inter-process communication, precedence graphs, critical section problem, semaphores, classical problems of synchronization. Deadlock problem, deadlock prevention, avoidance, recovery .

Memory Management: Concepts of memory management, logical and physical address space, swapping, contiguous and non-contiguous allocation, paging, segmentation, and paging combined with segmentation. Virtual memory, demand paging, page replacement algorithms, allocation of frames, thrashing, demand segmentation. Security threads protection intruders-Viruses-trusted system.

Introduction to Open Source technology, Files System hierarchy, Logging in, Simple commands like ls, cp, mv, wc, sort, tsort, cat, cut, grep, dd, head, tail, uniq, diff, echo, touch, which, whereis, whatis, type, who, whoami, finger, w (option and variations included), tty, ,uname, printf, ps, pwd , history, exec, kill, pkill, clear, lpstate, cancel, compress, uncompress, exit. Directory commands like: Brief introduction to file system, mkdir, dir, cd, df, dfspace, du, ll, dirname, rmdir, dir access permission, changing access permission for files and directories like: chmod, chgrp, chown, hard & soft links. Environments and path setting. I/O redirection & piping commands

vi editor: General startup of vi editor and it modes , Creating and editing files, features of vi, screen movement , cursor movement, insertion, deletion, searching, submitting operations, yank, put, delete commands, reading & writing files, exrc file for setting parameters, advance editing techniques, vim (improved vi).

Shell: meaning and purpose of shell, introduction to types of shell. The command line, standard input and standard output, redirection, pipes, filters special characters for searching files and pathnames. Built-ins, functions, history, aliases, job control, file substitution, source code management - RCS and CVS. awk utility.

Features of Linux: Drawbacks of Linux, Components of Linux, Memory Management Subsystems, Linux Process and Thread Management, File Management System, Device Drivers.

Linux Commands and Utilities: Entering the Machine, User Names and Groups, Logging In, Correcting Typing Mistakes, Format of Linux Commands, Changing Your Password, Characters with Special Meanings, Linux Documentation, The File System, Current Directory, Looking at the Directory Contents, Absolute and Relative Pathnames, Some Linux Directories and Files.

Linux Utilities and Editor: Some Useful Commands, Permission Modes and Standard Files, Pipes, Filters and Redirection, Shell Scripts, Graphical User Interface, Editor.

UNIX System Administration: System Administration, Installing Linux, Choosing an Installation

The bottom of the page features several handwritten signatures and initials in black ink. From left to right, there is a signature that appears to be 'J.P.', followed by a signature that looks like 'Alond', then a signature that is partially obscured and possibly '2. R...', followed by a signature that looks like 'H.C.', and finally a signature that looks like 'K.S.'.

Method, Choosing an Installation Class, Pre-installation checks, Installation, Booting the System, Maintaining User Accounts, File Systems and Special Files, Backups and Restoration.

TEXT BOOK:

1. A. Silberschatz et.al.-Operating System Concepts , 6th Edition, John Wiley Inc., 2003
2. H.M. Deitel -Operating Systems , 6th Edition, Pearson Education, 2006
3. A. Robbins- Linux Programming by Example- Pearson Education, New Delhi- 2005
4. Sumitabh Das : Your UNIX The Ultimate Guide; TMH

REFERENCE BOOKS:

1. D.M. Dhandhare - Operating Systems, 2nd Edition, Tata McGraw Hill, New Delhi, 2006
2. J.Goerzen- Linux Programming Bible, IDG Books, New Delhi- 2001
3. N.Mathew & R.Stones- Beginning Linux Programming Wiley Publishing India, 2004.
4. S.E. Mandnick & J.J. Donovan : Operating System; TMH

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